Checking the structural descriptions of finite hexagonal extensions

Human Compression project

Abstract

This supplement addresses structural group assertions in the retained source on hexagonal extensions. Exact marked presentations and incidence computations do not by themselves identify an asserted extension. We use their already certified permutation images to exhibit normal subgroups, identify quotients and check the indicated splittings. No presentation enumeration or geometric census is repeated here. Thirty-four printed shapes are verified under explicit brackets. One split-extension assertion is disproved and replaced by a certified nonsplit shape.

1 What must be proved

The source explicitly adopts the group notation of the Atlas (physical page 3). A string containing a colon asserts a split extension; a product sign asserts a direct product. Those assertions remain mathematical obligations even when the action is unnamed. In the certificates below an iterated expression is fully parenthesized. Unless parentheses in the source dictate otherwise, extensions are read from the left; every direct product is explicitly bracketed. We prove the resulting shape and do not claim it uniquely determines the group among all abstract groups of that shape.

The ambient group is not chosen by its order. We use the saved marked rotation group of the same literal source row. For a chiral row that is the full type-preserving group. For a regular row we use the previously checked inversion automorphism α and its faithful action on two copies: g acts as $(g, \alpha(g))$ and the new involution exchanges the copies. Its order is twice the rotation-group order. The earlier universal-presentation and geometry certificates establish that this is the required group; they are retained whole as premises of this supplement.

Lemma 1 (Structural witnesses)**.** *Let G be a finite group. Suppose $N \lhd G$, with N of a certified shape A and G/N of a certified shape B. Then G has extension shape $A.B$. If moreover a subgroup $C \leq G$ satisfies*

$$|N \cap C| = 1, \qquad |N||C| = |G|,$$

then $G = N \rtimes C$ and $C \cong G/N$. If also $[N, C] = 1$, then $G = N \times C$. These tests can be applied recursively to all the parenthesized factors.

Proof. The quotient homomorphism has kernel N and image G/N. Its restriction to C is injective. The order identity makes it surjective and gives $NC = G$. Thus conjugation by C supplies the action in the semidirect product. In the commuting case multiplication $N \times C \to G$ is a bijective homomorphism. Recursion applies these facts to the indicated subgroups and quotient groups, rather than inferring the factors from orders. □

Named simple and symmetric factors are checked by actual bijective homomorphisms to the declared standard groups. An elementary factor p^a is checked to be elementary abelian of order p^a. A factor p^{a+b} used here is certified with

$$\mathrm{Z}(P) = [P, P] = \mathrm{Frat}(P), \quad |\mathrm{Z}(P)| = p^a, \quad |P/\mathrm{Z}(P)| = p^b,$$

with both central subgroup and quotient elementary abelian. When $a = 1$ this proves that P is extraspecial. The exponent and actual centre, derived and Frattini subgroups are retained in the witness, so no stronger choice of extraspecial type is inferred from an unsigned source symbol. For Q_8 we check noncommutativity, order eight and exactly one involution; the two nonabelian groups of order eight are then distinguished directly. Cyclic factors are checked cyclic, not merely assigned by their order.

2 Finite implementation and trust

The required program reconstructs the previously accepted group and searches normal subgroups of each prescribed order. A candidate is accepted only after the entire kernel and quotient shapes are checked. For a colon it then obtains an actual complement and independently checks the two order identities in Lemma 1. For a direct product it also checks commutation on generators. The program retains the subgroup and complement generators, not just a printed abstract description.

The normal-subgroup search can first pass to a smaller quotient. If the required quotient has derived length at most d, its kernel contains $G^{(d)}$. If its solvable radical is trivial, the kernel contains the solvable radical of G. A normal p-subgroup lies in $O_p(G)$, so a p-power kernel can instead be found among the subgroups of that core which are normal in G. These are necessary conditions, not order-based identifications; all the asserted witness properties are checked afterward.

For index two, a complement is an involution outside the kernel. It is enough to examine a Sylow 2-subgroup, because conjugation preserves membership in a normal subgroup. For an S_3 quotient, a complement has a subgroup of order three. Conjugate this into a Sylow 3-subgroup and consider its conjugacy-class representatives there. For each representative y outside the kernel, an involution x inverting y must lie in the normalizer of $\langle y \rangle$. Conjugate it into a Sylow 2-subgroup of that normalizer. That conjugation sends y to y or y^{-1} and preserves the required relation. Thus these finite searches cover every possible complement. A direct-product complement lies in $C_G(N)$; it is a complement of $Z(N)$ there, provided $NC_G(N) = G$.

Normal-subgroup, quotient, Frattini, complement and homomorphism computations use GAP's exact finite-group algorithms. This is a declared computational premise, not a sample argument. A failed or timed-out search is not counted as a structural disproof. The certificate applies only to its exact row and literal presentation. The seven parameter-dependent or shared L121 rows are being treated in a separate supplement, and are not inferred from these finite rows.

3 The correction in Table 4, row 8

The source prints $2^{5+6} : A_5 \times 3 : S_3 : S_3$ for the row with both parameters $(2, 2)$, of full order 13 271 040. Its typography was checked directly on physical page 12. Every expression built from these factors using only split extensions and direct products contains a subgroup isomorphic to its A_5 factor, regardless of the ambiguous association. The certified group does not contain such a subgroup.

Indeed its derived series has orders

$$13\,271\,040, \quad 3\,317\,760, \quad 368\,640, \quad 122\,880, \quad 1\,920, \quad 120.$$

The final subgroup P satisfies $P' = P$. Every perfect subgroup K of G lies in every term of the derived series: from $K \leq G^{(j)}$ we obtain $K = K' \leq G^{(j+1)}$. Hence any A_5 subgroup would lie in P and have index two there. An index-two subgroup is normal and gives a nontrivial abelian quotient of P, contradicting $P' = P$. As a second exact obstruction, the complete

enumeration of the 120 elements of P gives exactly one involution; A_5 has fifteen. The recorded order histogram is

$$(1:1),\ (2:1),\ (3:20),\ (4:30),\ (5:24),\ (6:20),\ (10:24).$$

This is a finite certificate for the particular exact group, not a failure of a search for complements.

The positive corrected shape, with all association specified, is

$$\big((2^{5+6}.(A_5 \times C_3)) : S_3\big) : S_3. \tag{1}$$

The program exhibits a normal special subgroup of order 2^{11}, identifies the indicated quotient $A_5 \times C_3$, and gives actual complements at both outer colons. The inner dotted extension is nonsplit, since a complement would again contain A_5. The order, exact marked presentation and geometry of this source row are unchanged. We do not replace the printed expression silently or claim a new historical result.

4 Complete finite-row ledger

The table gives the exact bracketed assertions certified here. A dot records an extension; a colon records the actual complement checks. Only T4.8 is a replacement of the source's split claim. The p^{a+b} notation has the explicit special-group meaning stated above.

Row	Certified shape
T3.1	$(((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2) : C_2)$
T3.2	$(((C_2 \,.\, A_7) : C_2) : C_2)$
T3.5	$((Q_8 \,.\, (\mathrm{PSL}(2,7) \times C_2)) : S_3)$
T4.3	$(((3^{1+2} \times C_3) : S_3) : S_3)$
T4.5	$((3^{1+2} : C_2) : S_3)$
T4.6	$(((2^{1+6} : 3^2) : S_3) : S_3)$
T4.7	$(3^{1+2} : 2^2)$
T4.8	$(((2^{5+6} \,.\, (A_5 \times C_3)) : S_3) : S_3)$ (corrected)
T4.12	$((C_2 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T4.14	$((C_3 \,.\, \mathrm{PSU}(3,5)) : S_3)$
T4.15	$((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2)$
T4.16	$((C_2 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T4.17	$((C_2 \,.\, \mathrm{PSL}(2,7)) : C_2)$
T5.3	$(2^{1+6} : (A_5 : C_2))$
T5.5	$(((C_6 : S_3) : S_3) : S_3)$
T5.6	$(((2^{1+6} : C_3) : S_3) : S_3)$
T5.7	$((((2^{1+6} : C_6) : S_3) : S_3) : S_3)$
T5.9	$(((((S_3 \times C_3) : S_3) : S_3) : S_3) : S_3)$
T5.11	$((3^{1+2} : C_6) : S_3)$
T5.12	$((3^2 \times S_3) : S_3)$
T5.13	$(((3^{1+2} \times C_3) : S_3) : S_3)$
T5.14	$((((S_3 \times C_3) : S_3) : S_3) : S_3)$
T5.15	$(((C_4 \,.\, (2^4 : A_5)) : C_2) \times C_2)$
T5.19	$((Q_8 \,.\, \mathrm{PSL}(2,7)) : S_3)$
T5.21	$(((3^{2+6} : C_7) : C_3) : C_2)$
T6.1	$((2^{1+4} \times C_2) : S_3)$
T6.7	$((2^4 : S_3) : S_3)$
T6.9	$(((2^6 : C_7) : C_3) : C_2)$
T7.2	$((((2^4 : A_5) : C_2) : C_2) : S_3)$
T7.5	$((2^9 \,.\, \mathrm{PSL}(2,8)) : C_3)$
T7.7	$(((2^{1+4} : C_3) : S_3) : S_3)$
T8.4	$((((2^{1+6} : S_3) : S_3) : S_3) : S_3)$
T8.6	$(((2^{1+6} : C_3) : S_3) : S_3)$

Row	Certified shape
T8.7	$((((2^{1+6} : S_3) : S_3) : S_3) : S_3)$
T8.8	$(((C_2 \,.\, (A_5 \times (A_5 \times A_5))) : C_2) : S_3)$

5 Source and attribution boundary

The source is M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, the exact retained 21-page 2018 edition. The extension descriptions themselves are source assertions. The new work supplies their omitted explicit witnesses using the earlier saved images; it does not claim their historical discovery. The two previously proved order discrepancies, the local-period qualifications and all other source dispositions remain unchanged. The bounded prior review accepts only the exact presentations, geometry and previously verified named groups. It is not retroactively presented as acceptance of the structures checked here. There is no new Lean theorem, question solution or original credit in this supplement.